

# **Different Measures of Association Between Two Random Variables**

Literature Review by:  
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## *Abstract*

This study is to analyze various measures of association and more specifically question their merits and demerits in comparison to the basic standard measures. The goal of this literature review is to increase knowledge regarding effective measures of association in terms of their usage which could in certain ways improve the construct and development of many statistical applications.

## *Introduction*

Association measures are used to quantify the extent of a relationship between two variables. The first measure of association based on chi-square was Pearson's *Coefficient of Contingency*. An alternative to it, developed as an attempt to achieve a more appropriately normed measure of association was Cramer's *V*. Over time, we have witnessed the emergence of various generalized measures based upon many distributions, at times simple and at times even a tad bit complicated. In this data-driven age, the applications of these measures are extremely diverse. Not only in statistical theories but also in psychology, economics, and many more areas of research these measures provide a strong foundation for further development.

We are mainly interested in the computing degree of the statistical association between variables for the following reasons:

1. A weak association suggests the possibility that the association is a fluke that will not be replicated.
2. Changes in the degree of association may discern and describe important social trends or differences between populations.
3. The degree of association may help determine the relative importance of one variable that influences an outcome in comparison to all other influences.
4. The degree of association may reflect the degree to which two variables have a common cause.
5. The degree of association will indicate the utility of associated factors in predicting an outcome of interest.

## *Criteria of measures for association*

As a general guideline, we prefer to use a measure that is simple, easily interpretable and meaningful to the purpose and data available. Having a simple asymptotic theory under the hypothesis of independence is another another desirable criterion [13]. It, however, becomes a tedious job to choose from a pool of measures and create a proper ranking among them satisfying every criterion in the guideline stated.

Measures of association in sociological research, for instance, should be operationally interpretable in terms of the Proportional-reduction-in-error of estimation made possible by the relationship. There are various common measures which are proved to PRE measures whereas there is quite an abundance of measures which are not easy to interpret as the PRE measures. Probable consequences of utilizing proportional-reduction-in-error (PRE) measures in sociology include focusing attention on the form as well as the degree of association, as well as simplifying the choice and clarifying the interpretation of association measures. [1]

There exist 4 scales of data measurement, i.e. nominal, ordinal, interval, and ratio. Continuous variables is an umbrella term for Interval and ratio scales. It simplifies our process of decision making, based on the type of data available for analysis, to choose between measures according to their interpretation and ease. We present here a list of measures under the following 3 classes and point out their advantages, disadvantages, similarities (if any) as part of the literature review.

## 1. Nominal Variables:

Nominal variables are used to label values and there is no such ordering possible. For example 1 for female and 0 for male. It does not imply female is superior to male we can instead give 0 for female and 1 for male. Instead, we can also use 1 and 2 instead of 0 and 1. The following are the measures of association which help to determine the strength of association between nominal variables:

- (a) Phi ( $\phi$ ) Coefficient
- (b) Cramer's  $V$
- (c) Goodman and Kruskal's tau ( $\tau$ )
- (d) Goodman and Kruskal's lambda ( $\lambda$ )

Some points to be kept in mind before studying the measure of association of nominal type variables is

- Range of measure of association of nominal type variables is  $[0,1]$ .
- Value of 0 means no relationship and value of 1 means perfect relationship.

### (a) **Phi( $\phi$ ) Coefficient**

This measure is based on *chi square coefficient*. In this, we use the chi square value and the total number of observations  $N$  and obtain a coefficient called phi coefficient. The formula for its computation is:

$$\phi = \sqrt{\frac{\chi^2}{N}}$$

A value of 0.1 is *generally* considered to be small relationship, 0.3 a moderate relationship and 0.5 large relationship.

### **Advantages:**

- It is not influenced by the sample size provided proportion of cases in each group are similar.
- It is a symmetric measure.

### **Disadvantage:**

It is appropriate only for  $2 \times 2$ . If it is used for other dimension tables then it can take value greater than 1 so it becomes of no use.

### **Application:**

There has been use of this in the identification and ecological interpretation of various community types.[8] Besides, its application in the ecological field is quite widespread.

(b) **Cramer's V**

This measure is also based on *chi square coefficient*. It is an improvement over Phi Coefficient. The formula for computing Cramer's V is:

$$V = \sqrt{\frac{\chi^2}{N \times \min(r-1, c-1)}}$$

where,  $r = \text{the number of rows}$ ,  $c = \text{the number of columns}$

For a 2\*2 table, Cramer's V has same value as that of phi coefficient. We found some developments over Cramer's V ; it is stated as V' (more of an extension of V). [9]

**Advantages:**

It takes into account dimension of the table. It's also a symmetric measure.

(c) **Goodman and Kruskal's tau ( $\tau$ )**

This measure is based on *Proportional reduction in error measures*. It is interpreted as percentage reduction in prediction error when we know information about other (independent) variable. For example if Proportional Reduction in error is 0.2 it means prediction error is minimised by 20% when we are acquainted with independent variable. It requires variables to be classified as independent and dependent variable. The general formula for  $\tau$  is as follows:

$$\tau = \frac{E_1 - E_2}{E_1}$$

where,

$E_1 = \text{number of error without knowledge of independent variable}$

$E_2 = \text{number of error with knowledge of independent variable}$

We can infer that  $\tau$  depends on marginal distribution of independent variable.

Goodman and Kruskal's tau is equivalent to the correlation ratio, suitable for unordered categorical data. [10]

(d) **Goodman and Kruskal's Lambda( $\lambda$ )**

This measure is based on *Proportional reduction in error measures*. It is also interpreted in similar way as  $\tau$ . The difference is that instead of marginal distribution it depends on mode.

The general formula for  $\lambda$  is as follows:

$$\lambda = \frac{\epsilon_1 - \epsilon_2}{\epsilon_1}$$

where,

$\epsilon_1 = \text{number of errors using mode of dependent variable}$

$\epsilon_2 = \text{number of errors using mode of dependent variable by level of independent variable}$

**Advantage:**

- Values obtained by these measures can be directly interpreted unlike measures

based on chi square coefficient.

- It is easy to calculate and can be calculated with the help of softwares like SPSS, R with few clicks and few lines of code.

**Disadvantage:**

In this we need to figure out which variable is dependent and which is independent.

**Application:**

It has been found to be useful in many a fields of research , from social , psychological to genetics.It has also been used to find association in the medicinal world.  
[11]

## 2. Ordinal Variables:

When there is ordering or ranking associated with the different categories of the random variable, then it is called an ordinal variable. The categories of such a variable can be ranked from highest to lowest or lowest to highest. The distances or differences between these categories are uneven even though the difference between the labels are the same.

For example, 5 players participate in race competition and we rank them in order of their winning i.e. rank 1 to the player who came first and so on. When we compare the difference in the winning time of rank 1 and rank 2 with the difference in the winning time of rank 3 and rank 4, it will be different (or uneven) whereas the difference between the labels (ranks) is the same.

We can make use of the ranks of the categories to determine whether there is a positive or negative relationship between two random variables.

The following are the measures of association which help to determine the strength of association between ordinal variables:

- (a) Goodman and Kruskals' Gamma( $\gamma$ )
- (b) Kendall's Tau b ( $\tau_b$ ) and Kendall's Tau c ( $\tau_c$ )
- (c) Somers'  $D$

We'll first look at what are concordant and discordant pairs of observations before learning about these measures.

### Concordant and Discordant pairs

The pairs of observations for which the rankings are consistent are called concordant pairs whereas for which the rankings are inconsistent are called discordant pairs. In other words when the rankings of two observations on different variables are in the same direction then the pair is called a concordant pair. And when the rankings of two observations on different variables are in the opposite direction then the pair is called a discordant pair.

Consider the following example. Take two observations let's say  $Ob_1$  and  $Ob_2$  and have 2 variables X and Y with 5 categories each.

Suppose  $Ob_1$  is ranked 2 on variable X and 1 on variable Y while  $Ob_2$  is ranked 3 on variable X and 4 on variable Y, then  $Ob_1$  and  $Ob_2$  is a concordant pair as  $Ob_1$  is ranked 'low' on both the variables X and Y while  $Ob_2$  is ranked 'high' on both the variables. Now if  $Ob_1$  is ranked 2 on variable X and 5 on variable Y while  $Ob_2$  is ranked 4 on variable X and 1 on variable Y, then  $Ob_1$  and  $Ob_2$  is a discordant pair as  $Ob_1$  is ranked low on variable X and high on variable Y while  $Ob_2$  is ranked high on variable X and low on variable Y.

Consider the example with two variables

X: grades of the student in a subject with three levels namely "A", "B", "C" and

Y: interest of the student in the exam with two levels namely "Interested", "Not interested"

The cross-tabulated data is as follows:

| <b>X/Y</b> | <b>INTERESTED (High)</b> | <b>NOT INTERESTED (Low)</b> |
|------------|--------------------------|-----------------------------|
| A          | 12                       | 2                           |
| B          | 8                        | 2                           |
| C          | 2                        | 14                          |

We'll now determine the concordant and discordant pairs of observations for the above data. Each cell of the table contains the number of observations or cases corresponding to each pair of levels.

#### *Concordant pairs*

(High, Grade A)\*(Low, Grade B) =  $12 \times 6 = 72$  This is a concordant pair because the ranks of 12 observations (cell 1) are high on both variables X and Y and the ranks of 6 observations (cell 4) are low on both the variables. So, there will be a total of 72 (12X6) pairs. (Level 1, Grade A)\*(Level 2, Grade A) does not account for a concordant pair because the level for variable X i.e. Grade A remains the same for the observations from both the cells. Such cases account for ties.

Similarly taking into consideration one cell at a time, the concordant pairs are:

(High, Grade A)\*(Low, Grade B) =  $12 \times 6 = 72$  (High, Grade A)\*(Low, Grade C) =  $12 \times 14 = 168$  (High, Grade B)\*(Low, Grade C) =  $8 \times 14 = 112$  So the total number of concordant pairs are  $72 + 168 + 112 = 352$ .

#### *Discordant pairs*

(Low, Grade A)\*(High, Grade B) =  $2 \times 8 = 16$

This is a discordant pair because the ranks of 2 observations (cell 2) are high on variable X and low on variable Y and the ranks of 8 observations (cell 3) are low on variable X and high on variable Y. So, there will be a total of 16 (2\*8) pairs. (Low, Grade A)\*(Low, Grade B) does not account for a discordant pair because the level for variable Y i.e. Low remains the same for the observations from both the cells. Such cases account for ties.

Similarly taking into consideration one cell at a time, the concordant pairs are: (Level 2, Grade A)\*(Level 1, Grade B) =  $2 \times 8 = 16$  (Level 2, Grade A)\*(Level 1, Grade C) =  $2 \times 2 = 4$  (Level 2, Grade B)\*(Level 1, Grade C) =  $6 \times 2 = 12$  So the total number of discordant pairs are  $16 + 4 + 12 = 32$ .

#### (a) **Goodman and Kruskals' Gamma $\gamma$**

It measures the strength of association between two random variables both measured at ordinal levels and the data is presented in a bivariate table. It is asymmetric PRE measure so its value is not altered even if we don't specify the dependent and independent variable. Its value lies between +1 to -1 and if the value is 0 then it indicates the absence of association or in other words there is no relationship between the variables. To measure the significance of the Goodman and Kruskal's Gamma we generally use chi-square distribution, though there are some contradictory thoughts on the use of chi-square. [2]

The formula for computing gamma is:

$$\gamma = \frac{C - D}{C + D}$$



where,

C is the number of concordant pairs

D is the number of discordant pairs.

Using the above formula, we can draw the following conclusions:

- The value of gamma will be +1 only when there will be no discordant pairs indicating a perfect positive relationship.
- The value of gamma will be -1 only when there will be no concordant pairs indicating perfect negative relationship.

**Disadvantages:**

This measure does not take into account the ties and so leads to overestimating the strength of association between two ordinal variables.

**Applications:**

There has been quite a lot of use of this measure in various fields mainly biology [3] and various sociological researches (some of them are mentioned in [4]).

(b) **Kendall's Tau b( $\tau_b$ ) and Tau c( $\tau_c$ )**

These non-parametric measures also make use of the concordant and discordant pairs of observation but it also captures the additional information from the bi-variate table i.e. the number of tied observations on both the dependent and the independent variable. It is also an asymmetric measure whose value lies between +1 and -1.

The formula for computing Kendall tau b is:

$$\tau_b = \frac{C - D}{\sqrt{[N(N-1)/2 - T_x][N(N-1)/2 - T_y]}}$$

$$\tau_c = \frac{C - D}{\frac{1}{2}N^2[(m-1)/m]}$$

where,

C is the number of concordant pairs

D is the number of discordant pairs

N is the sum of the observations in the table

$T_x$  is the number of tied observations on the independent variable

$T_y$  is the number of tied observations on the dependent variable.

$m = \min(r, c)$  where c is the number of columns and r is the number of rows

If we represent the independent variable on the Rows and dependent on the Columns:

Then,  $T_x = \frac{(\sum \text{Rowsum} * (\text{Rowsum} - 1))}{2}$

And,  $T_y = \frac{(\sum \text{Rowsum} * (\text{Rowsum} - 1))}{2}$

The following are the **differences** between Tau b and Tau c:

- Tau b is used only when the number of rows is equal to the number of columns ( $r = c$ ) and in such a case the value of tau c is close to tau b.

- Tau b is a PRE (Proportional Reduction in Error) measure but tau c is not.

### **Comparison:**

- These measures are a good alternative to Spearman Rank correlation when the sample size is small and the data contains too many tied ranks.
- The value obtained by this measure is somewhat smaller than the Spearman  $\rho$  correlation.
- Kendall correlation is more efficient than the Spearman correlation as it has a smaller asymptotic variance.

Moreover, there has been quite a development in the estimation of Kendall's tau under censoring. [7]

### **Applications:**

It too used in wide range of fields mainly in the socio-economic matter as well as biological subjects [5]. There is also an adaptive use of this to test against serial dependence in a univariate time series context. [6]

### **(c) Somers' D:**

This measure is very much similar to Kendall's tau measures but the only difference is that it takes into consideration the ties on only the dependent variable. It is an asymmetric measure as its value changes depending on which variable is identified as a dependent variable. It is a PRE measure so it can be interpreted as an improvement in predicting the dependent variable by knowing the independent variable.

The formula for computing Somers' D is:

$$d = (C - D)/(C + D + T_y)$$

Where  $T_y$  is the number of tied observations on the dependent variable.

### **Comparison:**

- $d$  will always have a lower value than gamma as the formula for Somers'  $D$  is equivalent to that for gamma with an adjustment for the ties on the dependent variable in the denominator.
- It is considered to be a better measure than tau c as it has a PRE interpretation.

### **Applications:**

There has been quite and extensive use of this measure in the real-life applications. It was used in banking when there was a major change in banking structure of Florida in 1980s. [12]

### 3. Continuous Variables:

They are also known as quantitative variables. Continuous variables are further categorised into 2 categories, Interval and Ratio variables. Interval variables are those which have order and difference between two values is meaningful. However, true zero can't be defined on such a scale. Ratio variables have all the properties of interval variables, except that they do have an absolute zero. This allows for a wide range of both descriptive and inferential statistics to be applied over ratio scale. We here would like to first mention some widely used measures of association in this category, in order to later draw comparisons between few other measures against these.

- **Pearson correlation coefficient**( $\rho$ ) is a measure of **linear** relationship between two variables and is one of the most widely used measures of correlation. However it is sensitive to outliers because it uses arithmetic mean in its computation. It also assumes normality of the population, which may not always be the case. It also assumes data to be homoscedastic in nature.

- **Spearman correlation coefficient** can be used when the assumptions of the Pearson correlation are violated. It is a non-parametric test which determines the strength and direction of the **monotonic** relationship between two variables.

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

where  $d_i$  = difference in paired ranks and number of cases.  
It's more robust to outliers than Pearson.

On a side note, we'd like to mention that in the case when sample size is small, or in the case of multiple ties in the data, we can use Kendall's tau measure of association. Similar to Spearman's rho, Kendall's tau also measures **monotonic** relationship. The following is a list of measures which are applicable to continuous random variables based on data type.

#### (a) **Point Biserial Correlation Coefficient**

This type of correlation measures the strength of association when one data is continuous and other is dichotomous. There are some **assumptions** which are required to hold:

- (i) Continuous variables should be approximately normally distributed for each category of the dichotomous variable.
- (ii) There should be no outliers for the continuous variable for each category of the dichotomous variable.
- (iii) The continuous variable should have equal variances.

The formula is

$$r_{pb} = \frac{M_1 - M_0}{s_n} \sqrt{\frac{n_1 n_0}{n(n-1)}}$$

where  $M_1$  is the mean value 11 on the continuous variable X for all data points in group 1 of variable Y

$M_0$  is the mean value on the continuous variable X for all data points in group 2

of variable Y.

$n_1$  is the number of data points in group 1,

$n_0$  is the number of data points in group 2 and  $n$  is the total sample size.

- Point biserial ranges from 0 to 1, where if  $r_{pb}$  is 0, it means there is no relationship between X and Y and if  $r_{pb}$  is 1, There exists a perfect relationship.

- If there is positive correlation .Point Biserial correlation coefficient measures monotonic relationship, it means that as one variable increases, the other variables also increase . For Example, daily temperature and no of sales of ice-cream per day and vice-versa.

#### **Disadvantages**

- One of the drawback for this correlation is the basic assumption of correlation analysis that the variables be randomly independent.

- Also,that it requires no outliers for continuous data thus making it sensitive to outliers.

#### **Applications**

- Out of many applications of Point Biserial correlation, [21] talks about its application in Item Analysis.

Item analysis is a procedure where items on a test are reviewed, both qualitatively and statistically, in order to verify if they all meet minimum quality-control criteria.

#### **(b) Biserial Correlation Coefficient**

Its similar to Point Biserial but here the dichotomous variable should be fundamentally normally distributed .For example, depression level can be measured on a continuous scale, but can be classified dichotomously as high/low.It is denoted by  $r_b$  and the formula is

$$r_b = \frac{Y_1 - Y_0 * (pq/Y)}{\sigma_y}$$

where

$Y_0$  = mean score for data pairs for  $x=0$ ,

$Y_1$  = mean score for data pairs for  $x=1$ ,

$q$  = proportion of data pairs for  $x=0$ ,

$p$  = proportion of data pairs for  $x=1$ ,

$\sigma_y$  = population standard. deviation.

#### **(c) Tetrachoric Correlation Coefficient**

This type of correlation is appropriate when we want to measure correlation between 2 dichotomous data whose underlying distribution is bivariate normal.It is denoted by  $r_t$ . For a 2\*2 table of frequencies with the assumption of bi-variate normality

| <b>X/Y</b> | 0   | 1   | Total |
|------------|-----|-----|-------|
| 1          | a   | b   | a+b   |
| 0          | c   | d   | c+d   |
| TOTAL      | a+c | b+d | n     |

The coefficient is formulated as follows:

$$r_t = \cos \left\{ \frac{180}{1 + \sqrt{\frac{bc}{ad}}} \right\}$$

- $r_t$  lies between -1 and 1.
- Tetrachoric correlation does not have its range affected by unequal data marginals and its significance can be assessed using appropriate tables.

### **Applications:**

This measure is used to measure default correlation of firms [16] and it is also used in forecast evaluation or verification of quantitative forecasts [17].

#### **(d) Intra-class Correlation Coefficient**

While Pearson is inter-class and used for validity of variables, Intra-class correlation describes how strongly units in the same group resemble each other. There are many forms of ICC each with a different purpose. The intra-class correlation is also defined for data sets with groups having more than 2 values. For  $n=3$ , the formula is

$$r = \frac{\sum_{n=1}^N [(x_{n,1} - \bar{x})(x_{n,2} - \bar{x}) + (x_{n,1} - \bar{x})(x_{n,3} - \bar{x}) + (x_{n,2} - \bar{x})(x_{n,3} - \bar{x})]}{3Ns^2}$$

where

$$\bar{x} = \frac{\sum_{n=1}^N (x_{n,1} + x_{n,2} + x_{n,3})}{3N}$$

and

$$s^2 = \frac{\sum_{j=1}^3 \sum_{n=1}^N [(x_{n,j} - \bar{x})^2]}{3N}$$

- A high Intra-class Correlation Coefficient (ICC) close to 1 indicates high similarity between values from the same group.
  - A low ICC close to zero means that values from the same group are not similar.
- When just restricting it to two variables, the intra-class divides their variance into two parts. One is the variance of each case on the two variables and is called the within-class variance. If the magnitudes for each case on the two variables are the same, this variance is zero. The second part is the variance of the variable across the cases, which is the between-class variance. The intra-class correlation is then simply the between-class minus the within-class variance as a ratio to the sum of the two kinds of variance. And the significance of this ratio can be assessed.

### **Applications:**

ICC is majorly used in large data sets to find dependence between 2 or more variables. One of them being, The application of Intra-Class Correlation Coefficient (ICC) in assessing the reliability of medical instruments measuring continuous outcomes[22]

Apart from the above mentioned correlations, there are other measures of association between random variables each for a different purpose and use irrelevant of the scale of measurement of data. While some coefficients focus on testing independence (Hoeffding's Test for independence)[24], some measure the amount of information (Mutual information). Some statisticians came up with unified measure which have properties of both Pearson and Spearman [23] while some are entirely new based on different methods such as entropy based, copulas based , kernel based and list goes on. In this literature review, we will focus on 3 such measures of association, Distance Correlation, Maximal Information Coefficient and another new measure of correlation proposed by Sourav Chatterjee.

(a) **Distance Correlation**[18]

Analogous to Pearson product-moment correlation, Distance correlation is introduced in 2005 by J.Szekely which can measure the dependence between two random vectors of 'arbitrary, not necessarily equal dimension. Distance correlation provides a new approach to the problem of testing the joint independence of random vectors.

Let  $X \in \mathbb{R}^p$  and  $Y \in \mathbb{R}^q$  where  $p$  and  $q$  are positive integers. let the characteristic functions of  $X$  and  $Y$  be  $f_X$  and  $f_Y$  respectively and the joint distribution  $f_{XY}$ . Then define the Distance co-variance  $\mathcal{V}$  as

$$\begin{aligned}\mathcal{V}^2 &= ||f_{XY}(t, s) - f_X(t)f_Y(s)||^2 \\ &= \frac{1}{c_p c_q} \int_{\mathbb{R}^{p+q}} \frac{|f_{XY}(t, s) - f_X(t)f_Y(s)|^2}{|t|_p^{1+p} |s|_q^{1+q}} dt ds\end{aligned}$$

where  $||\cdot||$  is the norm. to test the hypothesis,  $H_0 : f_{XY} = f_X f_Y$  vs.  $H_1 : f_{XY} \neq f_X f_Y$  Distance correlation (dCor)  $\mathcal{R}(X, Y)$  is defined as

$$\mathcal{R}^2(X, Y) = \begin{cases} \frac{\mathcal{V}^2(X, Y)}{\sqrt{\mathcal{V}^2(X)\mathcal{V}^2(Y)}} & \text{if } \mathcal{V}^2(X)\mathcal{V}^2(Y) \geq 0 \\ 0 & \text{if } \mathcal{V}^2(X)\mathcal{V}^2(Y) = 0 \end{cases} \quad (1)$$

**Properties:**

By definition, Distance Correlation (dCor) is analogous in some respects to the corresponding Pearson product-moment correlation.

By analogy, certain properties of classical correlation and variance definitions should also hold for dCor and dVar.

Properties of dCor:

- $0 \leq \mathcal{R}_n \leq 1$ .
- If  $\mathcal{R}_n(\mathbf{X}, \mathbf{Y}) = 1$ , then there exist a vector  $\mathbf{a}$ , a nonzero real number  $b$  and an orthogonal matrix  $C$  such that  $\mathbf{Y} = \mathbf{a} + b\mathbf{X}C$ .
- If  $E(|X|_p + |Y|_q) < \infty$ , then  $0 \leq \mathcal{R}_n \leq 1$ , and  $\mathcal{R}_n(X, Y) = 0$  if and only if  $X$  and  $Y$  are independent.

Properties of dVar:

For r.v with finite first moments,

- i.  $dVar(X) = 0$  implies that  $X = \mathbb{E}[X]$ , almost surely.
- ii.  $dVar(a + bCX) = |b|dVar(X)$  for all constant vectors  $a$  in  $\mathbb{R}^p$ , scalars  $b$  and  $p \times p$  orthonormal matrices  $C$ .
- iii.  $dVar(X + Y) \leq dVar(X) + dVar(Y)$  for independent random vectors  $X \in \mathbb{R}^p$  and  $Y \in \mathbb{R}^q$

**Asymptotic Properties:-**

If  $E(|X|_p + |Y|_q) < \infty$ , then

- 1) When  $X$  and  $Y$  are independent and

$$n\mathcal{V}_n^2/S_2 \xrightarrow[D]{n \rightarrow \infty} \mathcal{Q},$$

where  $\mathcal{Q}$  is a non negative quadratic form of centered Gaussian random variables and  $\mathbb{E}[\mathcal{Q}] = 1$ .

- (ii) If  $X$  and  $Y$  are dependent, then

$$n\mathcal{V}_n^2/S_2 \xrightarrow[P]{n \rightarrow \infty} \infty,$$

**Advantages:**

(Since its analogous to Pearson , i am comparing it to it) • While Pearson is applicable to variables of same dimension, distance correlation coefficient can be applied to variables of any dimension.

- Pearson correlation coefficient can detect only linear association but  $\mathcal{R}$  is capable of detecting nonlinear associations .Hence, the distance correlation coefficient provides more information than the Pearson coefficient.
- Unlike the Pearson coefficient, the distance correlation coefficient is zero if and only if the variables are independent.

**Disadvantages:**

- Pearson coefficient can be estimated with higher accuracy than the distance correlation coefficient.

**Applications:**

- This measure is widely used for discovering association between large astro-physical variables which have non-linear relationship between them[19].

(b) **Maximal Information Coefficient**

Many economic and biological process variables in general tend to deviate from a linear dependence. Maximal Information Coefficient helps capture the extent of dependence in various functional and non functional relationships. MIC lies between  $[0,1]$ . Value of the coefficient is close to 1 if there exists any of the possible relationships, such as linear, cubic, sinusoidal, parabolic, exponential, etc. This property is called equitability of MIC. value close to 0 reflects closeness to statistical independence. It is formulated as such:

$$MIC(X, Y) = \max \left( \frac{I(X, Y)}{\log_2 \min(n_X, n_Y)} \right)$$

where, *mutual information*  $I(X, Y) = \mathbb{E} \log_2 \left( \frac{f(X, Y)}{f(X)f(Y)} \right)$  and  $n_X, n_Y$  are no. of observations available for X, Y respectively.

**Advantages:**

- It is robust to outliers, and does not make assumptions about the distribution of random variables.
- Of the other measures available in literature, it is most 'general', in terms that it can detect any kind of dependence among variables.

**Disadvantages:**

- It's computationally challenging.
- Value between 0 and 1, makes interpretation easy, but fails to provide the direction of dependence.

**On comparison** of Pearson correlation coefficient with Distance correlation and Maximal Information Coefficients for a large sample data set, relation between Pearson and Distance correlation is 'sharper', against that in Pearson and MIC score. Also, the Distance correlation can be proven to be stronger measure of association when compared to MIC, for variables in astrophysical database. [20]

There exists a package in python, "mictools", which has inbuilt functions to compute MIC. For R programming, the package available is called *minerva*: Maximal Information-Based Non-parametric Exploration R package for Variable Analysis.

(c) **Sourav's  $\xi$ : New coefficient to measure the dependence** [13]

Sourav Chatterjee has put forward a new measure which satisfies most of the criterion that a desirable measure of association should possess.

Let  $(X, Y)$  be a pair of random variables, where Y is not a constant. Let  $(X_1, Y_1), \dots, (X_n, Y_n)$  be i.i.d. pairs with the same law as  $(X, Y)$ , where  $n \geq 2$ .

- *In case of no ties*

Rearrange the data as  $(X_1, Y_1), \dots, (X_n, Y_n)$  such that  $X_1 \leq \dots \leq X_n$ . Let  $r_i$  be the rank of  $Y_i$ , that is, the number of  $j$  such that  $Y_j \leq Y_i$ . The new correlation coefficient is defined as

$$\xi_n(X, Y) = 1 - \frac{3 \sum_{i=1}^{n-1} |r_{i+1} - r_i|}{n^2 - 1}$$

- *In case of ties*

Choose an increasing rearrangement as above by breaking ties uniformly at random. Let  $r_i$  be as before, and let  $l_i$  to be the number of  $j$  such that  $Y_j \geq Y_i$ . Then the formula is

$$\xi_n(X, Y) = 1 - \frac{n \sum_{i=1}^{n-1} |r_{i+1} - r_i|}{2 \sum_{i=1}^n l_i (n - l_i)}$$

**Properties:-**

- $\xi_n$  is not symmetric. i.  $\xi_n(X, Y) \neq \xi_n(Y, X)$ . This is because we may want to understand if Y is a function of X. not if one of the variables is a function of other.
- $\xi_n$  is 0 when X and Y are independent and is 1 when at least one of X and Y



is a measurable function of other. Thus the range of  $\xi_n$  is  $[0,1]$

(iii) The law of  $X$  and  $Y$  can be discrete, continuous, light-tailed or even heavy-tailed but  $Y$  should not be constant.

(iv) The coefficient  $\xi_n(X, Y)$  remains unchanged if we apply strictly increasing transformations to  $X$  and  $Y$ , because it is based on ranks.

(v) If there are no ties among the  $Y_i$ 's, the maximum possible value of  $(X, Y)$  is  $\frac{(n-2)}{(n+1)}$ , which is attained if  $Y_i = X_i$  for all  $i$ .

### Asymptotic Properties:-

Here we give a brief overview of these properties without delving into the details and the proofs.

**Case:-** When  $X$  and  $Y$  are independent and  $Y$  is continuous

Under the hypothesis of independence and the assumption that  $Y$  is continuous.

When  $n \rightarrow \infty$

$$\sqrt{n}\xi_n(X, Y) \sim N(0, 2/5)$$

**Case:-** When  $X$  and  $Y$  are independent

Even when  $X$  and  $Y$  are independent but  $Y$  is not continuous, then also  $\xi_n$  converges in distribution to a centered Gaussian law, but the variance  $\tau^2$  has a more complicated expression. For each  $t \in \mathbb{R}$ , let  $F(t) = P(Y \leq t)$  and  $G(t) = P(Y \geq t)$ . Let  $\phi(y, y') = \min[F(y), F(y')]$ . Then

$$\tau^2 = \frac{\mathbb{E}\phi(Y_1, Y_2)^2 - 2\mathbb{E}(\phi(Y_1, Y_2)\phi(Y_1, Y_3)) + (\mathbb{E}\phi(Y_1, Y_2))^2}{(\mathbb{E}(G(Y)(1 - G(Y)))^2}$$

where  $Y_1, Y_2, Y_3$  are independent copies of  $Y$ .

### Advantages:

- The formula is easy to understand and interpret.
- Due to its simple formula, It can be computed very easily.
- Since it is based on ranks, Its robust to outliers and is in-variant under monotone transformations of the data.
- $\xi_n$  can actually measure the strength of relationship between 2 variables.
- It has simple asymptotic properties under the hypotheses of independence.
- The test of independence based on  $\xi_n$  being consistent against all alternatives, with no exceptions is its unique property.
- the formula can be used irrespective of law (i.e both categorical and continuous) except that  $Y$  should not be constant

### Disadvantages:

When the signal is smooth and non-oscillatory,  $\xi_n$  is less powerful than other tests.

## *Conclusion*

This Literature review begins with the splitting of random variables into 3 categories -nominal, ordinal and continuous variables and has proposed different measures of association within these variables along with their advantages and disadvantages. All these measures have their interpretation and applications. The problem with each of the measures was also discussed and answered with the help of the next following measure in most of the cases. We also discuss few measures generic to the type of data, and dig deep into mainly 3 new such measures, Distance Correlation, Maximal Information Coefficient, and a development made by Sourav Chatterjee.

This literature review also tries to touch the matter of monotonicity of the measures. Some of the monotone measures of association of ordinal or interval variables include Goodman and Kruskal's Gamma ( $\gamma$ ), Somers'  $D$ , Kendall's tau ( $\tau$ ), or Spearman's rho ( $\rho$ ) [14]. A three-facet model for an optimal selection of measures of association has already been developed and elaborated on [15].

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